Phase-Resolved Transparency Classification in Commodity Supply Chains:

A Structural Triadic Scoring Framework (TVPCI)

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Abstract

Opacity in commodity supply chains is commonly described in narrative or intent-based terms. This paper proposes a structural alternative: transparency is a property of *observable evidence attached to each phase of a chain*, and can be scored by a phase-resolved framework with three functionally distinct roles. We formalize the **Transparency via Phase-resolved Classification and Indexing (TVPCI)** framework, in which each phase $p \in \{0, \dots, P\}$ of a commodity chain hosts three non-negative observables: a **negotiation state** N_p (declared position and mass-balance coherence), a **definition/limitation score** D_p (bounding evidence: audit scope, policy constraints, counterparty opacity caps), and a **contribution score** C_p (independent corroboration depth: third-party assays, geospatial attestations, physical touchpoints). These roles are motivated by the tholonic triadic framework; here they organize indicators rather than recurrences.

A **phase-local balance functional** $B(D_p, C_p)$ penalizes one-sided evidence: strong disclosure without corroboration, or deep tracing without audit scope, both degrade the balance score. A **transition penalty** τ_p reduces the contribution of a phase when its N-state carries unresolved red flags from the prior phase. A weighted aggregate index combines these components.

The methodological contribution is a reusable schema: once an indicator dictionary is fixed for a commodity, the framework produces a reproducible, phase-consistent score for any observed chain. We demonstrate with a synthetic eight-phase gold ladder (phases 0 through 7, anchored at an exchange-linked endpoint) and a parallel synthetic shea-oil profile, chosen because both commodities have well-documented opacity concerns at extraction and aggregation stages. All figures use synthetic or schematic data; no proprietary or audited datasets are calibrated in this preprint. Limitations, including post-hoc flexibility and legal definition gaps, are stated explicitly.

Contents

1 Introduction

Commodity supply chains for minerals, agricultural products, and energy feedstocks increasingly attract regulatory scrutiny. The OECD Due Diligence Guidance for minerals from conflict-affected regions [9], the EU Conflict Minerals Regulation [3], and emerging mandatory human-rights due-diligence legislation in several jurisdictions all require evidence of responsible sourcing. Yet the practical bottleneck in applying these requirements is not regulatory intent but *measurement*: how does an auditor, exchange, or buyer convert a collection of documents, assay records, and counterparty attestations into a single, reproducible statement about chain transparency?

Existing approaches fall into three broad categories. First, binary pass/fail certification (conflict-free, Fairtrade) is legible but coarse; it typically reduces multi-phase evidence to a single threshold judgment that discards phase-specific variation. Second, narrative risk assessments (the dominant form in OECD guidance) are rich but poorly comparable across chains and not amenable to automated aggregation. Third, weighted indicator composites (as used in some ESG scoring) aggregate indicators without enforcing phase ordering or testing for evidence balance.

TVPCI addresses a gap that each of these approaches leaves open: a *phase-ordered, balance-tested* aggregate score. Three design principles distinguish it.

Phase ordering matters. Transparency evidence does not commute across phases. An assay record at refining is meaningful only if the chain upstream (aggregation, mine gate) has non-zero observability. A phase-resolved score must propagate gaps rather than allow a high-scoring phase to mask a low-scoring predecessor.

Evidence balance is structural. A chain that accumulates strong bounding (policy constraints, audit scope) but weak independent corroboration is structurally opaque in a different sense than one with deep corroboration but no audit limits. The balance functional $B(D, C)$ operationalizes this distinction.

Organization precedes prediction. TVPCI does not claim to predict enforcement outcomes, commodity prices, or fraud rates. It claims to organize available evidence into a consistent ordinal structure. Predictive applications require additional labeled data and pre-registered calibration; those are explicitly deferred.

The triadic roles (N, D, C) used here share labels with the tholonic mathematical framework developed in a companion manuscript [8]. In that paper, (N, D, C) are positions in convergent recurrences; here they are *classes of indicators* attached to a phase node. The analogy is deliberate: the functional distinction between a running state, a bounding constraint, and an integrating corroboration is preserved across domains. No mathematical results from the companion paper are imported here.

The remainder of the paper is organized as follows. §2 surveys related work. §3 develops the phase-resolved ontology for gold. §4 defines the triadic observables and indicator dictionary. §5 introduces the balance functional, transition penalties, and aggregate index. §6 presents a worked numerical example. §7 presents the synthetic two-commodity comparison and sensitivity analysis. §8 discusses limitations. §9 concludes.

2 Related work

2.1 Regulatory frameworks: OECD and conflict minerals

The OECD five-step framework [9] establishes strong company management systems, identifies and assesses risks, designs strategies to respond to identified risks, carries out third-party audits, and reports annually on supply chain due diligence. TVPCI is aligned with this framework but differs in focus: OECD guidance describes *processes* (what a company should do); TVPCI scores *evidence states* (what a chain observably demonstrates at each phase). TVPCI could in principle generate a phase-resolved audit artifact mapping to OECD step 2 (risk identification) and step 3 (strategy implementation monitoring).

The EU Conflict Minerals Regulation [3] requires importers of tin, tungsten, tantalum, and gold above threshold volumes to carry out OECD-aligned due diligence. It does not prescribe a numerical scoring method; TVPCI could serve as an audit-support layer.

2.2 GRI and sector reporting standards

The Global Reporting Initiative provides topic-specific standards for supply chain disclosures [4, 5]. These standards require disclosure of the proportion of screened suppliers but do not enforce phase consistency. A supplier that rates well on social criteria may still be embedded in a chain with opaque aggregation-phase intermediaries. TVPCI phase-local scoring exposes this gap explicitly.

2.3 Chain-of-custody models

Chain-of-custody certification schemes for forestry, aquaculture, and minerals track material through a defined custody chain but typically do so via claims rather than independent measurement at each link. TVPCI is compatible with chain-of-custody programs: existing custody claims feed into N_p (the state score), while the evidence supporting those claims determines D_p and C_p. A chain with custody claims but low C_p will produce low balance scores at those phases.

2.4 ESG composite indices

Quantitative ESG ratings aggregate many indicators but typically do not enforce sequential chain structure, phase ordering, or a balance test between complementary evidence types. Several studies have documented low inter-rater agreement among major ESG providers [2], attributable partly to scope and weighting differences. TVPCI fixed indicator dictionary and phase ordering constraints are designed to reduce such variation, at the cost of requiring commodity-specific up-front specification work.

2.5 Systems-theoretic frameworks

Holarchy [6], viable systems models [1], and autopoiesis theory each describe multi-level systems with internal role differentiation. These frameworks inform the design intuition behind the (N, D, C) roles but are not formally used in the scoring rules.

3 Phase-resolved ontology

3.1 Rationale for phase indexing

A supply chain is a directed acyclic graph of custody handoffs, physical transformations, and documentation events. For scoring purposes, we coarsen this graph into a linear sequence of **phases** by grouping nodes with similar observability characteristics and regulatory exposure. This coarsening trades resolution for tractability; the appropriate phase count is commodity-specific.

For gold, eight phases capture the dominant transformation events while remaining fine enough to localize opacity. A chain that aggregates phases 0 through 3 into a single opaque "upstream block would lose the ability to distinguish a transparent refinery from an opaque aggregator.

Phase skipping must be handled explicitly. If phase p is not observed, the model assigns a **skip penalty**: N_p inherits N_{p-1} deflated by a skip factor $\sigma \in (0, 1)$, and $D_p = C_p = 0$, producing $B_p = 100 \exp\big(-2 \max(D_{p-1}, C_{p-1})/(\max(D_{p-1}, C_{p-1}) + \varepsilon)\big)$, which collapses to near-zero when the prior phase had unequal observables. This is conservative by design: missing a phase is treated as maximum imbalance.

3.2 Gold phase ladder

Phase 7 is the **high-observability anchor**: exchange rules impose delivery specifications, and open-interest and warehouse-stock data are publicly reported. This makes phase 7 the most constrained node in the model: the most *measurable*.

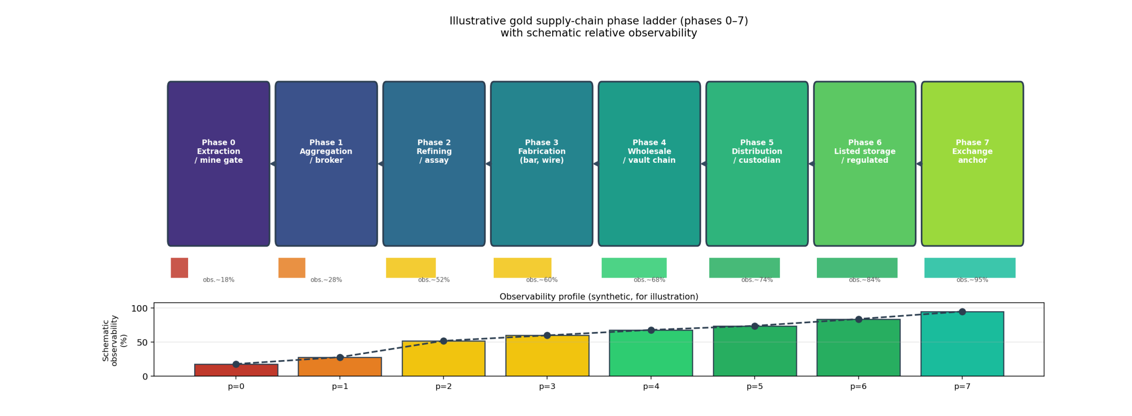

Figure 1: Phase-resolved gold ladder with synthetic observability profile. Each bar represents the aggregate TVPCI score at that phase under the synthetic gold baseline described in §7.

Table 1: Gold supply chain phases, with schematic labels and representative transformation or custody events.

Phase	Label	Transformation or custody event
0	Extraction / mine gate	Ore extraction, initial weighing, first lot numbering
1	Aggregation / broker	Consolidation of material, first documentary claim
2	Refining / assay	Smelting, cupellation, purity certification; lot identity re-established
3	Fabrication	Bar or wire production; serial marking; LBMA-eligible or equivalent
4	Wholesale / vault chain	Transfer to secure storage, custodian documentation
5	Distribution	Retail-linked holdings, commercial delivery chains
6	Listed storage	Regulated custodian, eligible-vault status, centralized reporting
7	Exchange anchor	Delivery-eligible bars under exchange rules

3.3 Generalization

The same machinery applies to any commodity once a phase table and indicator dictionary are fixed. Agricultural oil chains would replace metallurgical steps with harvest, processing, and blending phases. Battery minerals would follow analogous extraction-to-cell steps. The phase count need not be eight; chains with fewer distinct custody events may use four or five phases without loss of scoring validity.

4 Triadic observables and indicator dictionary

4.1 Role definitions

At each phase p, three non-negative scalars $(N_p, D_p, C_p) \in [0, 100]^3$ are computed by applying a **scoring rule** to raw indicator measurements.

N_p **(negotiation / state).** The declared chain position at phase p, evaluated for internal consistency. Indicators assess whether the custody claim entering phase p is coherent: mass-balance residuals, time-delta anomalies versus published benchmark lead times, identifier continuity, and document-count completeness.

D_p **(definition / limitation).** The extent to which the *boundaries* of the claim are defined. Indicators assess audit scope, counterparty documentation, policy existence and substance, legal definitional alignment, and sampling plan rigor.

C_p **(contribution / corroboration).** The depth and independence of evidence that *supports* the claim from external sources. Indicators assess third-party assay certificates, geospatial attestation, cross-system endorsements, physical touchpoints, and whistleblower or source-community corroboration.

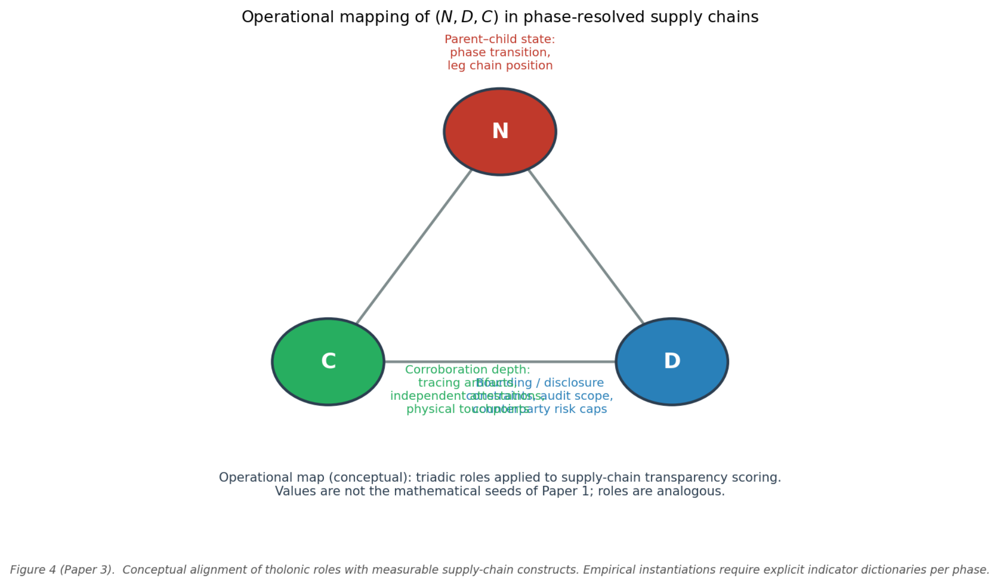

Figure 2: Conceptual map of the three indicator roles (N_p, D_p, C_p) at a supply chain phase node. N_p captures the declared state and its internal coherence; D_p bounds the claim; C_p corroborates it independently.

4.2 Specimen indicator dictionary (gold, phases 0 and 2)

An indicator dictionary fixes the mapping from raw evidence items to (N_p, D_p, C_p) values. The tables below provide a specimen for two phases; a complete deployment requires entries for all eight phases.

Phase 0 (Extraction):

Phase 2 (Refining):

Scoring rules are **additive up to a cap of 100**; the cap enforces the $[0, 100]$ range without requiring a precise probabilistic interpretation. Final scale and weights within each role must be pre-registered before data collection to limit researcher degrees of freedom.

Remark 4.1 (Role distinctness). Two indicators may share numerical values in a particular chain; what the triadic structure requires is that *families* of indicators remain functionally distinct. An operator can maximize D_p without any improvement to C_p (by writing stronger policies with no independent verification), and vice versa. The balance functional in §5 penalizes this one-sided optimization.

Table 2: Specimen indicators for Phase 0 (Extraction).

Indicator	Role	Scoring rule (schematic)
Mass-in / mass-out ratio within 2%	N	+20 if within tolerance
Lot number traceable to mining permit	N	+15 if traceable; +8 if partial
Written environmental / social policy exists	D	+20 if substantive; +8 if nominal
Counterparty identity verified (KYC)	D	+20 if documented
Independent field visit within 12 months	C	+25 if within window
Geospatial trace	C	+20 if available

Table 3: Specimen indicators for Phase 2 (Refining).

Indicator	Role	Scoring rule (schematic)
Input mass vs. output mass within 0.5%	N	+25 if within tolerance
Lot continuity from phase 1 to phase 2	N	+20 if continuous
LBMA GDL or equivalent listing [7]	D	+30 if listed
Published responsible sourcing policy	D	+15 if substantive
Third-party purity certificate	C	+30 if from accredited lab
Chain-of-custody cross-matched to registry	C	+20 if matched

5 Scoring model

5.1 Phase-local balance

Define the **balance score**

$$B_p \;=\; 100\,\exp\!\left(-\,\frac{2\,|D_p - C_p|}{\max(D_p,\,C_p)+\varepsilon}\right),$$

where $\varepsilon > 0$ is a small regularization constant. Properties: (i) $B_p = 100$ iff $D_p = C_p$; (ii) B_p is symmetric in D_p and C_p; (iii) for fixed ratio $r = D_p/C_p$, B_p depends only on the relative imbalance; (iv) $B_p \to 0$ as the imbalance ratio grows. The exponential form guarantees $B_p > 0$ for any finite observables. The penalty grows more steeply than linear for imbalance $|D - C|/\max(D, C) > 0.3$ while remaining forgiving near the diagonal.

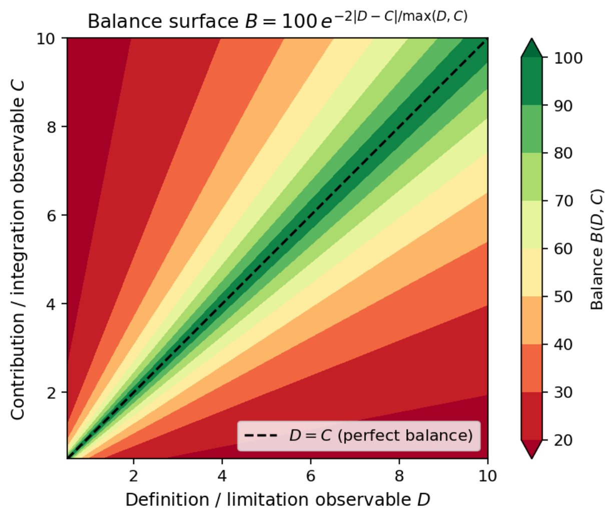

Figure 2 (Paper 3). Contours of the phase-local balance score. Along $D = C$, $B = 100$; as $|D - C|$ grows relative to $\max(D, C)$, B decays exponentially.

Figure 3: Balance surface $B(D, C)$ as a function of $(D_p, C_p) \in [0, 100]^2$. The $D = C$ diagonal achieves $B = 100$; scores fall symmetrically as imbalance grows.

5.2 State quality mapping

Let $g\colon [0, 100] \to [0, 100]$ map phase state N_p to a quality score. The simplest case is $g(x) = x$ (the identity). More complex mappings, such as threshold-sigmoid functions treating $N_p < 30$ as near-zero quality, are admissible and must be declared in advance. For all synthetic examples here, $g = \mathrm{id}$.

5.3 Transition penalty

Let $\mathrm{RF}(N_p, N_{p-1}) \geq 0$ be a **red-flag functional** that increases when the state claim entering phase p is inconsistent with the state leaving phase $p - 1$. Concrete instances: the absolute mass-balance discrepancy normalized to phase-$(p-1)$ output mass, or the count of identifier mismatches between consecutive phases. The **transition penalty** is

$$\tau_p \;=\; \tanh\!\big(\alpha \cdot \mathrm{RF}(N_p,\, N_{p-1})\big),$$

with $\alpha > 0$ scaling sensitivity. Since tanh is bounded, $\tau_p \in [0, 1)$. For the first phase, $\tau_0 = 0$.

5.4 Aggregate index

Let $w_p > 0$ with $\sum_{p=0}^{P} w_p = 1$ be phase weights. The **TVPCI aggregate** is

$$\mathrm{TVPCI} \;=\; \sum_{p=0}^{P} w_p \cdot \Big(\beta\, B_p + (1-\beta)\, g(N_p)\Big) \cdot \big(1 - \gamma\, \tau_p\big),$$

where $\beta \in (0, 1)$ weights balance against state quality and $\gamma \in (0, 1)$ scales the transition-penalty discount. The product form ensures a high penalty at phase p reduces that phase contribution without affecting other phases. The constants $(\beta, \gamma, \alpha, w_p)$ must be **declared ex ante**.

Monotonicity. If $D_p = C_p$ and $\tau_p = 0$ for all p, then TVPCI $= \sum_p w_p N_p$, the weighted average of state scores. Balance and penalty terms can only reduce the index; it is monotonically increasing in B_p, $g(N_p)$, and $(1 - \tau_p)$.

Range. TVPCI $\in [0, 100]$ since $B_p, g(N_p) \in [0, 100]$ and $1 - \gamma\tau_p \in (0, 1]$ for $\gamma \in [0, 1)$.

Parameter count. For $P+1$ phases the model has $P+3$ unconstrained free parameters plus the indicator dictionary. For $P = 7$ this gives ten free parameters, manageable if controlled through pre-registration and sensitivity reporting.

6 Worked numerical example

We apply the aggregate formula to three representative phases from the synthetic gold chain: phase 0 (extraction), phase 2 (refining), and phase 7 (exchange anchor). Parameters: $\beta = 0.5$, $\gamma = 0.5$, $\alpha = 1.0$; weights $w_0 = 0.20$, $w_2 = 0.35$, $w_7 = 0.45$ (downstream phases receive higher weight, reflecting greater commercial relevance).

Table 4: Worked three-phase TVPCI computation on the synthetic gold baseline. A full computation uses all eight phases.

Phase	D_p	C_p	N_p	B_p	RF_p	τ_p	w_p	Contribution
0	18	12	22	74.1	0.00	0.000	0.20	9.61
2	55	58	52	97.9	0.15	0.149	0.35	26.08
7	94	93	90	99.5	0.02	0.020	0.45	42.81
							TVPCI	**78.5**

Phase 0 contributes little because its observables are low (extraction is genuinely hard to observe) and its weight is lower. Phase 2 contributes substantially despite carrying a transition penalty from a small mass-balance anomaly. Phase 7 contributes most, reflecting both high local scores and dominant phase weight.

7 Synthetic comparison and sensitivity analysis

7.1 Two-commodity comparison

Synthetic profiles for gold and shea oil are constructed by choosing phase baselines that reflect qualitative differences in chain observability: gold benefits from standardized assay procedures and listed-exchange endpoints; shea oil has a less formalized upstream (largely artisanal collection) but well-documented refinery and cosmetics-industry quality standards. The shea profile is therefore lower at phases 0 and 1, converging toward (but not reaching) the gold profile by phase 7.

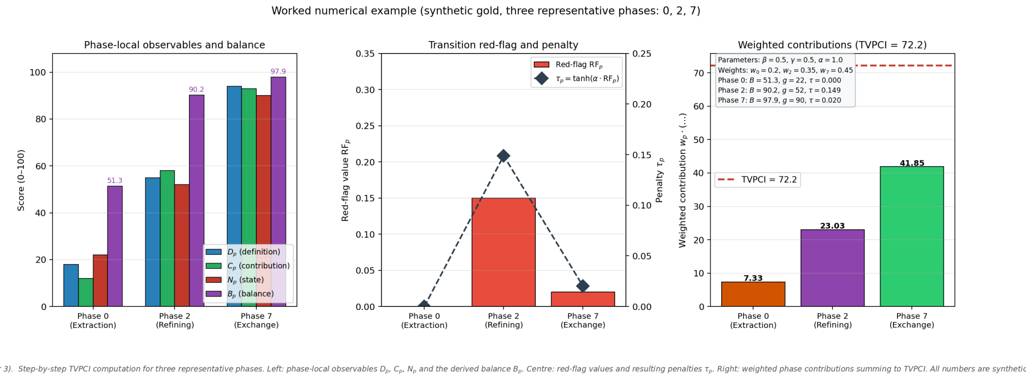

Figure 4: Step-by-step TVPCI computation for the three-phase synthetic example, showing the phase contributions and the aggregate score of 78.5.

Both profiles carry 90% bootstrap confidence intervals derived by adding normally distributed indicator noise to each phase baseline (standard deviations ranging from 2 to 7 points), resampled 2 000 times. These intervals illustrate *sensitivity to indicator measurement error*, not empirical sampling variability from a real dataset.

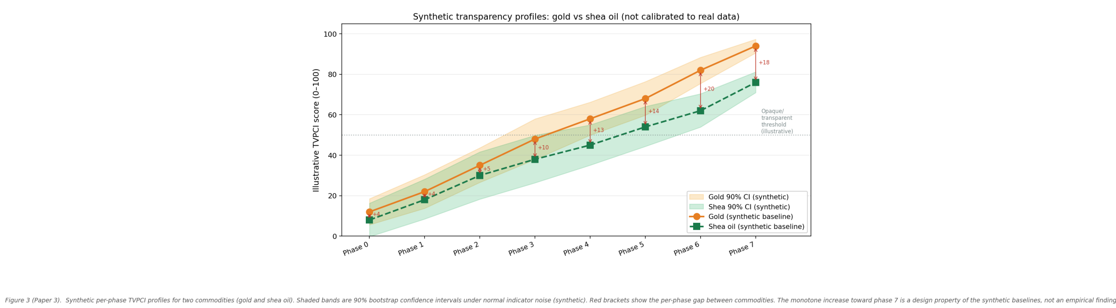

Figure 5: Synthetic TVPCI profiles for gold and shea oil with 90% bootstrap confidence intervals. The gold curve crosses the 50-point illustrative threshold at phase 2 (refining); the shea curve crosses it at phase 4. Intervals narrow at phase 7, reflecting lower assumed indicator noise at exchange-anchored endpoints.

Key observations: (i) Phase 1 shows the largest absolute gap between commodities (roughly 10 points) because artisanal shea collection has fewer standardized custody events than small-scale gold mining subject to conflict-minerals reporting; (ii) the gold curve crosses the 50-point illustrative threshold at phase 2 (refining), the shea curve at phase 4; (iii) both curves narrow in confidence interval at phase 7, reflecting the lower indicator noise assumed at exchange-anchored endpoints.

7.2 Sensitivity to β and γ

A well-calibrated TVPCI should not change its ordinal ranking of chains substantially when β and γ are varied within plausible ranges. The sensitivity surface below shows the aggregate score on the synthetic gold baseline as β varies from 0 to 1 and γ from 0 to 1.

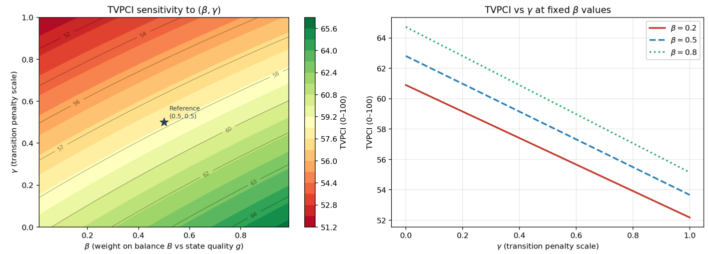

Figure 6: TVPCI sensitivity surface over hyperparameters (β, γ) on the synthetic gold baseline. Scores are monotonically decreasing in γ for all β. Maximum score variation across the full parameter space is approximately 8 points.

Observations: (i) the score is monotonically decreasing in γ for all β, confirming that stronger transition penalties uniformly reduce the aggregate; (ii) the score is nearly flat in β for this synthetic chain, because $B_p \approx g(N_p)$ across most phases; (iii) the maximum score variation across the full parameter space is approximately 8 points. For chains with larger imbalances, β sensitivity would be substantially higher. This motivates reporting the sensitivity surface alongside any published TVPCI calibration.

8 Discussion

8.1 Organizational vs. predictive claims

TVPCI is an *organizational* instrument: it specifies how to aggregate evidence into a phase-resolved score. It does not claim that a score above a threshold guarantees absence of fraud, conflict sourcing, or human-rights violations. Those predictive claims require labeled outcome data and prospective validation. Framing TVPCI as a scoring schema rather than a classifier avoids overstating its current evidential status.

8.2 Post-hoc flexibility and pre-registration

Any multi-parameter index with analyst discretion over weights and indicator definitions risks being optimized to produce a desired result. TVPCI addresses this through three mitigations. First, indicator dictionaries and all parameters must be frozen before observing chain data; any change must be documented as a model revision. Second, sensitivity surfaces over (β, γ) must be published alongside point estimates. Third, a held-out commodity provides a partial out-of-sample check; if the model was calibrated on gold, shea oil provides a structural generalization test.

8.3 Score stability and Lipschitz bounds

An index score is operationally useful only if small changes in indicators produce small changes in the aggregate. Let $\delta_p = (\delta N_p, \delta D_p, \delta C_p)$ be a perturbation at phase p with

$\|\delta_p\|_\infty \leq \epsilon$. Since B_p is Lipschitz in (D_p, C_p) (the exponential function has bounded derivative for bounded inputs) and g is Lipschitz by assumption, the aggregate changes by at most $O(P\epsilon)$ under small perturbations. Making this Lipschitz bound explicit would allow an auditor to certify that a score of, say, 72 is robust to indicator measurement error of ± 3 points.

8.4 Missing phases and phase skipping

Chains lacking documented evidence for an intermediate phase present a practical challenge. The proposed convention treats a skipped phase as having maximum imbalance ($B_p \approx 0$) and no state contribution. If there is positive reason to believe the phase was uneventful, a less aggressive default may be adopted, but must be pre-registered as a modeling choice.

8.5 Legal and jurisdictional definitions

Several indicators in the specimen dictionary refer to legal concepts (origin, conflict-affected area, KYC documentation) that are jurisdiction-specific. A global deployment of TVPCI would require jurisdiction-specific indicator dictionary variants. Standardization bodies such as OECD and LBMA could provide a mapping.

8.6 Regulatory alignment

Mapping TVPCI to OECD five-step guidance [9] is feasible. OECD step 2 (risk identification) corresponds to identifying phases with low N_p or low B_p; step 3 (strategy response) to the action required to raise specific phase scores; step 4 (third-party audit) to independently verifying the C_p indicators. A companion compliance note mapping TVPCI fields to OECD steps and GRI 308/414 disclosures [4, 5] would be a useful practical deliverable.

9 Conclusion

We introduced **TVPCI**, a phase-resolved transparency scoring framework for commodity supply chains. The framework assigns three functionally distinct observables (N_p, D_p, C_p) to each phase, computes a balance score B_p penalizing one-sided evidence, and applies transition penalties when chain-state claims are internally inconsistent across phases. A weighted aggregate combines these elements into a single index on $[0, 100]$.

The framework was demonstrated on synthetic eight-phase gold and shea-oil profiles. Worked numerical examples, sensitivity analysis over β and γ, and a two-commodity comparison illustrate the scoring machinery. All presented data are synthetic.

Immediate next steps for an empirical deployment are:

1. Fix indicator dictionaries for one commodity through expert consultation and pre-registration.

2. Apply the schema to real chains with known audit outcomes, using those outcomes as a validation signal.

3. Publish the sensitivity surface and held-out commodity results alongside any point estimates.

4. Develop a Lipschitz stability certificate for the chosen indicator-noise level.

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A Figure checklist

Table 5: All figure assets for this paper. Filenames begin with `2_`, reserved for this paper.

File	Section	Content
`2_gold-phase-ladder.png`	§3.2	Eight-phase gold ladder
`2_balance-metric-B.png`	§5	Balance surface $B(D, C)$
`2_ndc-operational-map.png`	§4	Indicator-role map
`2_synthetic-transparency-by-phase.png`	§7	Two-commodity profiles
`2_sensitivity-surface.png`	§7	Hyperparameter surface